Kinetic Energy Driven Ferromagnetic Insulator

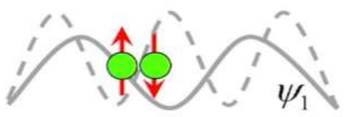
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Abstract

We construct a minimal model of interacting fermions establishing a ferromagnetic insulating phase. It is based on the Hubbard model on a trimerized triangular lattice in the regime of $t \gg |t'|$ with $t > 0$ and t', the intra- and inter-trimer hopping amplitudes respectively. At the $1/3$-filling, each trimer becomes a triplet spin-1 moment, and the inter-trimer superexchange is ferromagnetic with $J = -\frac{2}{27}\frac{t'^2}{t}$ in the limit of $U/t = +\infty$. As U/t becomes finite, the antiferromagnetic superexchange competes with the ferromagnetic one. The system enters into a frustrated antiferromagnetic insulator when $U_F/t > U/t \gg 1$ where$U_F/t \sim 10$. In contrast, a similar analysis performed on the trimerized Kagome lattice shows that only antiferromagnetic superexchange exits at 1/3-filling. The effect of threading flux is also studied.

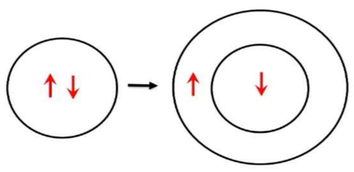
The quantum origin of ferromagnetism

- Driving force: (direct) **exchange interaction** among electrons with parallel spins



$$E_{\uparrow\uparrow} < E_{\uparrow\downarrow}$$

- **Stoner criterion:**



$$UN_0 > 1$$

U – average interaction strength;
N_0 – density of the states at Fermi level

Nagaoka's ferromagnetism

Hamiltonian:

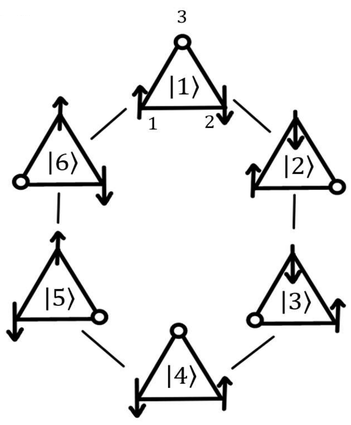
$$H = t\sum_{\langle ij\rangle}\left(c_{i\sigma}^{+}c_{j\sigma} + h.c.\right) + U\sum_{i} n_{i\uparrow}n_{i\downarrow}$$

1) Almost half-filling with exactly one hole
2) $U \to \infty, t > 0$
3)Transitivity condition

The GS is non-degenerate and fully polarized – Ferromagnetism occurs !

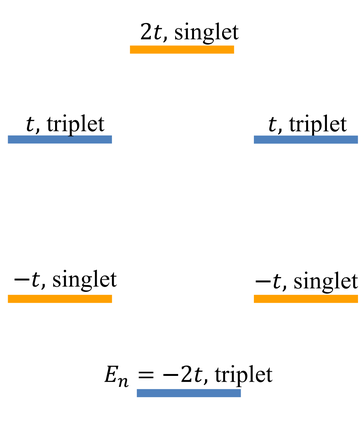
Toy model (trimer)

Configuration space
$S_z = 0$ sector



Diagonalization

Momentum space (spectrum)



Spectrum: $E_n = -2t\cos\left(n\frac{\pi}{3}\right), n = 0, \pm1, \pm2, 3$
Eigen-states: $|k_n\rangle \propto \sum_{i=1}^{6} e^{ik_n i}|i\rangle, k_n = n\frac{\pi}{3}$

$|GS\rangle \propto$ (|1⟩ + |4⟩)

+ other two possible permutations

which is a triplet – the basic unit of FM !

Reference

1. Phys. Rev. Lett. 62, 1201 (1989)
2. Commun. Math Phys. 158, 341 (1993)
3. Phys. Rev. Lett. **75**, 4678(1995)
4. Phys. Rev. Lett. **112**, 217201(2013)
5. Phys. Rev. X 5, 021032(2015)

Trimerized triangular lattice (TTL)

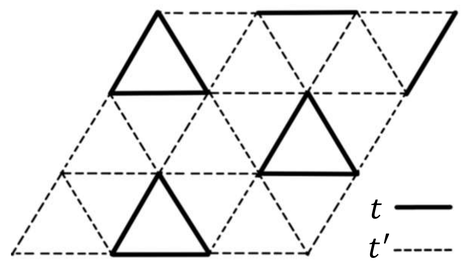
$$H = H_0 + H'$$

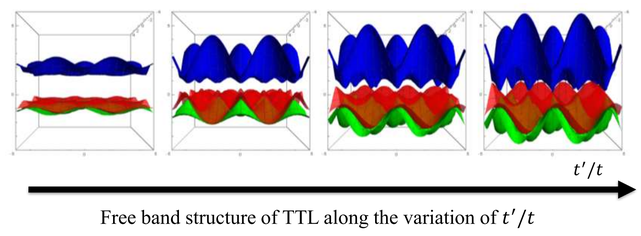
$$H_0 = t\sum_{\langle ij\rangle}\left(c_{i\sigma}^{+}c_{j\sigma} + h.c.\right) + U\sum_{i} n_{i\uparrow}n_{i\downarrow}$$

$$H' = t'\sum_{\langle\langle i'j'\rangle\rangle}\left(c_{i'\sigma}^{+}c_{j'\sigma} + h.c.\right)$$

$\langle ij\rangle$ is the intra-trimer bonds
$\langle\langle ij\rangle\rangle$ is the inter-timer bonds

$t > 0$ is important for establishing FM





Free band structure of TTL along the variation of t'/t

Effective FM Heisenberg model

- An analogy between the model and Heisenberg model is established by :

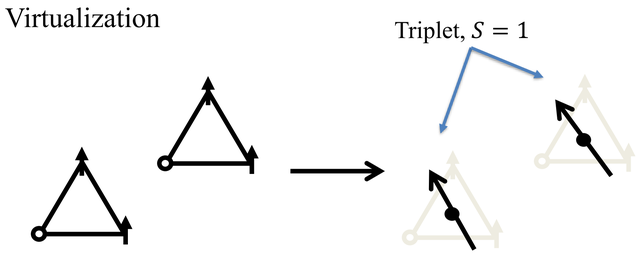
$$H = H_0 + H'$$
$$\Rightarrow H_{eff} = J\sum_{\langle ij\rangle}\vec{S}_i\cdot\vec{S}_j + Const$$

➢ According to SU(2) symmetry, the Hilbert space is separated into three sectors:
$$1 \otimes 1 = 2 \oplus 1 \oplus 0$$
➢ The energy gain of effective Heisenberg model is easy to calculate, which are $-J, J, 2J$
➢ The energy gain of original model could be calculated by perturbation theory as $t' \ll t$ is assumed, which are $-\frac{10t'^2}{27t}, -\frac{6t'^2}{27t}, -\frac{4t'^2}{27t}$

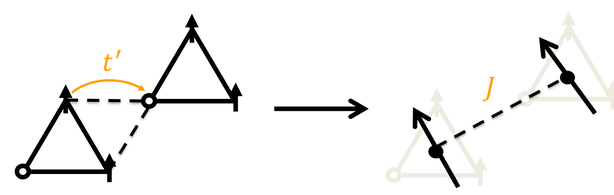
- The effective coupling is determined as

$$J = -\frac{2t'^2}{27t} < 0$$

Virtualization



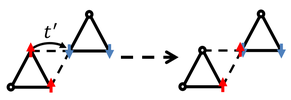
After introducing inter-hoppings



- FM occurs when
  1. 1/3-filling
  2. $U \to \infty$
  3. $t' \ll t$

Interaction-tuning FM-AFM transition

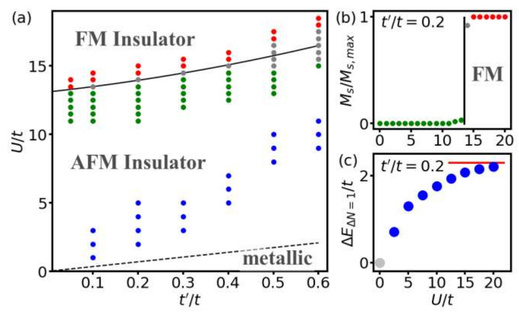
When $t \ll U < \infty$, intermediate states with double occupancy matters



2nd perturbation theory gives:

$$J = -\underbrace{\frac{2}{27}\frac{t'^2}{t}}_{\text{FM channel}} + \underbrace{\frac{62}{81}\frac{t'^2}{U}}_{\text{AFM channel}}$$

DMRG simulation:



Conclusion

- We discover a FM insulating state, which is established in the trimerized triangular lattice at the 1/3-filling with $t > 0$ in the regime of $U \gg t \gg |t'|$.
- The inter-trimer hoppings generate superexchange couplings to lower the kinetic energy.
- At $U/t \to \infty$, only the FM superexchange exists.
- As U/t becomes finite, both the FM and AFM superexchanges contribute, and the former wins over the latter around $U/t \gtrsim 15$.

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